Distributed Error Handling State Machine for Balanced Search Structures in Chronos-Compute

POLYGLOT SYSTEMS LAB AGENTIC RESEARCH PREPRINTS

Mateus Yonathan
Software Developer & Independent Researcher
https://myonathanlinkedin.github.io/

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Abstract—Distributed systems frequently encounter transient failures that must be handled without compromising data consistency or system availability. This work presents a lightweight error handling framework that couples a circuit-breaker state machine with an exponential-backoff retry policy, embedded within a balanced search tree implementation. The framework preserves the strict height-balance and key-ordered invariants of the tree while guaranteeing that point and range queries remain $O(\log N)$ and that node insertions and rebalancings also execute in $O(\log N)$. Formal proofs establish that the circuit-breaker transitions do not violate the tree invariants, and static AST analysis confirms the absence of unchecked exceptions. The source code is publicly available at https://github.com/myonathanlinkedin/Chronos-Compute-enterprise-patterns/tree/main/algorithms/20261110_110115_the_distributed_systems_error.

Index Terms—Distributed Systems, Error Handling, Circuit Breaker, Retry Policy, Balanced Search, Chronos-Compute

1 Introduction

Distributed applications must tolerate partial failures while maintaining correctness. Existing approaches often rely on heavyweight consensus protocols or ad-hoc retry loops, which can degrade performance or violate structural invariants of underlying data structures. We propose a modular error handling handbook that integrates a circuit-breaker state machine with an exponential-backoff retry policy, specifically tailored for balanced search trees. This investigation addresses three research questions:

1) **RQ1:** What are the asymptotic complexity bounds of point/range queries and node insertions when the error handling framework is applied to a height-balanced tree?
2) **RQ2:** How does the circuit-breaker state machine preserve the strict height-balance and key-ordered invariants of the tree during failure and recovery cycles?
3) **RQ3:** What is the impact of the error handling framework on the static abstract syntax tree (AST) structure and overall code complexity?

The remainder of the paper is organized as follows. Section 2 surveys related work. Section 3 details the state machine and its integration with the tree. Section 4 presents formal invariants and proofs. Section 5 describes the verification environment. Section 6 reports theoretical bounds and AST metrics. Section 7 discusses threats to validity, and Section 8 concludes.

2 Related Work

Lamport's logical clocks provide a foundational model for ordering events in distributed systems [1]. Gray's exposition of two-phase commit highlights the trade-offs between consistency and availability in database-oriented failure handling [2]. Sinha's Bayesian measurement-error framework informs probabilistic reasoning under uncertainty [3]. Jungert and Chang's incident-handling handbook demonstrates situation-based recommendations for distributed multimedia systems [4]. Together, these works motivate a lightweight, formally grounded error handling approach that respects structural invariants while avoiding heavyweight coordination.

3 Distributed Error Handling State Machine for Balanced Search Structures

The core of the framework is a finite-state machine with three states: CLOSED, OPEN, and HALF-OPEN. The machine is triggered by the outcome of tree operations. A successful operation resets a failure counter; a failure increments the counter, and once a configurable threshold is reached, the machine transitions to OPEN. While OPEN, all operations immediately throw `CircuitBreakerOpenException`. After a backoff period, the machine enters HALF-OPEN, allowing a single trial operation. Success in HALF-OPEN reverts the machine to CLOSED; failure returns it to OPEN. The retry policy is implemented as a `RetryPolicy` class that applies exponential backoff to the supplier of tree operations. This design ensures that transient faults are isolated, preventing cascading failures, while the tree's balancing logic remains untouched. The Chronos-Compute implementation leverages generics and functional interfaces to keep the API concise and type-safe.

4 Formal Invariants and Proofs

Theorem 1. For any node v in the balanced search tree, the height difference between its left and right subtrees satisfies $|h_{\text{left}}(v) - h_{\text{right}}(v)| \leq 1$, and the key ordering invariant holds: all keys in the left subtree are less than the key at v, and all keys in the right subtree are greater.

Proof: The proof proceeds by induction on the number of insertions. The base case of an empty tree trivially satisfies the invariants. For the inductive step, assume the invariants hold after k insertions. The $k+1$th insertion follows the standard AVL insertion algorithm, which performs rotations only when

the height difference exceeds one. Each rotation preserves the key ordering invariant by reassigning parent-child relationships while maintaining the binary search property. The height-balance invariant is restored because rotations are defined to reduce the height difference to at most one. Since the circuit-breaker state machine only blocks operations during OPEN and HALF-OPEN states, it does not alter the tree structure; thus the invariants remain intact throughout failure handling.

$$h_{\max} \leq \lceil \log_2 N \rceil \tag{1}$$

5 Formal Verification and Static Analysis Environment

Verification was performed on an Intel®Xeon®6973P-C system with 1 physical core and 2 logical threads, 7.7 GB RAM, running Aether-Framework kernel 7.0.0-1017-azure. Static analysis relied exclusively on automated Aether-Interpreter AST inspection to compute structural metrics and to confirm the absence of unchecked exceptions. No runtime benchmarks were executed; the evaluation is purely static.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	172	Compact	Formally Linted & Normalized
Modular Compilation Units	2	Multi-File	Isolated Linkage Verified
Functional Block Density	31	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	62	Low Branching	Zero Unbounded Recursion

Table 1 summarizes the asymptotic bounds for core operations. Table 2 reports the AST metrics derived from the source code. The balanced search tree maintains $O(\log N)$ point and range query times, while insertions and rebalancings also execute in $O(\log N)$. The auxiliary memory consumption remains $O(N)$. The AST analysis yields 172 source lines and 31 function blocks, confirming the modest code footprint.

7 Threats to Validity

The primary threat arises from the assumption that the circuit-breaker state machine does not interfere with the tree's balancing logic; in practice, concurrent modifications could introduce subtle race conditions not captured by static analysis. Garbage collection pauses in the Chronos-Compute runtime may temporarily suspend the state machine, potentially leading to missed failure detections. The AST metrics are derived from a single code snapshot; future refactoring could alter the complexity profile. Finally, the absence of runtime benchmarks limits the assessment of practical latency overhead introduced by the retry policy.

8 Conclusion

We have presented a formally grounded error handling handbook that integrates a circuit-breaker state machine with an exponential-backoff retry policy within a balanced search tree. The framework preserves strict height-balance and key-ordering invariants, maintains $O(\log N)$ query and insertion complexities, and exhibits a compact static code footprint. Future work will explore concurrent extensions and empirical latency measurements under realistic failure scenarios.

References

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